

# JULIAN ZHANG

+1 (647) 914-2970 | [julian.zhang@uwaterloo.ca](mailto:julian.zhang@uwaterloo.ca) | [linkedin.com/in/zhaju](https://www.linkedin.com/in/zhaju) | [zhaju.github.io](https://zhaju.github.io)

## EDUCATION

### University of Waterloo

(Sep 2024 - Apr 2028)

Bachelor of Computer Science and Finance Double Major (CFM)

*President's Scholarship of Distinction*

## EXPERIENCE

### RBC Investor Services | Software Engineering Intern

(May 2025 - Aug 2025)

- Used **Python** to automate financial reporting processes responsible for **\$1.2 billion** in transaction value
- Built predictive models using **pandas** and **Yahoo Finance** to forecast institutional client trading patterns compared to the S&P500 and TSX60 indices. Presented work to upper-level executives
- Engineered interactive dashboards serving **500+** Canadian and **100+** international clients

### Oakville Centre For Vision | Software Engineer

(May 2024 - Sep 2024)

- Used **C# MonoGame** and **Python** to create a web app for eye-tracking goggles. Used it to analyze vision patterns in professional athletes, aiding the clinical practice of optometrist Dr. Ronald Gall
- Focused on data analysis of **8+ tests** that examined Vector3s for saccades, fixations, and pursuits
- Collaborated with a hardware team to modernize a **9,000+ line** codebase for improved functionality

### University of Toronto Clean Energy Lab | AI Research Assistant

(Jun 2022 - Sep 2023)

- Applied Crystal Graph Convolutional Neural Network (CGCNN) to examine trends in chemical properties
- Analyzed results for **material discovery** of lithium-ion alternatives for sustainable energy production
- Worked with Prof. Oleksandr Voznyy to implement ML models, achieving a **97%** classification accuracy

### Anthropic | Campus Ambassador

(Aug 2025 - Dec 2025)

- Founded the University of Waterloo's **Claude Builder Club**, hosted **5+ events** for **600+** members
- Held workshops explaining key AI concepts (APIs, Agents, Model-Context Protocol (MCP), Alignment)
- Converted **200+** Claude signups across Waterloo through unique **in-person and online** marketing
- Led the launch of **Claude 4.5 Sonnet** on campus, hosted a competition generating **7,000+** impressions

## PROJECTS

### SignWave | OpenCV, Flask, MediaPipe, Whisper API

- An AI translator that converts text and audio into American Sign Language animations, improving accessibility for the deaf and hard of hearing. **1st Place and Best Use of GitHub @ JamHacks VII**

### EdgeFund | Yahoo Finance, Pandas, NumPy, matplotlib

- A digital stockbroker that uses statistical methods (Sharpe Ratio, beta) to select a portfolio and optimizes weights with **Monte Carlo Simulations**. Outperforms the S&P/TSX by **30% over 1 year**

## SKILLS

- **Programming Languages:** C/C++, C#, Python, Java, JavaScript, HTML/CSS, Racket/Scheme
- **Frameworks & Libraries:** React, Node, NumPy, matplotlib, Pandas, PyTorch, TensorFlow, scikit-learn
- **Tools & Technologies:** Git/GitHub, Docker, Linux, Jupyter, Excel/VBA, Power BI, PostgreSQL
- **Certifications:** Supervised ML, Advanced Learning Algorithms, Unsupervised Learning

## AWARDS

**Best Debater in Canada** | Canadian National Debate Championship

(Nov 2023)